

1. DEFINITION

DEFINITION	
	Glycerol dibehenate EP Glyceryl dibehenate NF Glyceryl behenate ChP

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Glyceryl Behenate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	R8WTH25YS2
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Glycerol and Behenic acid. The esterification step is followed by atomization using a spray-cooling technique. No solvent or catalyst used during the manufacturing process.

COMPOSITION
Well defined multi-constituent substance composed of mono-, di- and triglycerides of Behenic acid (C22), the diester fraction being predominant (Specification: 40-60%).



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Full Labeling (Key letters system)	
$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Glyceryl Behenate	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	

4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	4663
Canada MF (Type III – Excipient)	2015-156
For submission of the DMF in another country, please contact your local Gattefossé representative	

5. PHARMACOPeias

CONFORMITY TO PHARMACOPeIAS	
Reference	Conforms to monograph
European Pharmacopoeia (Ph.Eur.) Current edition	Glycerol dibehenate
USP/ NF Current edition	Glyceryl dibehenate
Japanese Pharmaceutical Excipient JPE Current edition	/
Chinese Pharmacopoeia Current edition	Glyceryl behenate



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6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓	✓(5) GRAS 184.1328		
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

7. SPECIFIC PHARMACEUTICAL REGULATIONS

CHINA BUNDLING REVIEW			
(Generic name) Chinese name	Registration N°	Company	CFDA/CDE Reference (http://www.cde.org.cn/yfb.do?method=main)
(Glyceryl behenate) 山嵛酸甘油酯	F20170000058 (Route 1)	Gattefossé SAS	Raw material approved for use in marketed formulations in Joint-Review with drug applications (Status A)
(Glyceryl behenate) 山嵛酸甘油酯	F20190000560 (Route 4)	Gattefossé SAS	Raw material has not been approved yet for use in marketed formulations in Joint-Review with drug applications. However, the excipient dossier has already been submitted (Status I).
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Glyceryl behenate (or glycerol dibehenate)	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines. Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines.	In Over The Counter products: Glyceryl behenate is i) permitted for use as an oral excipient; ii) permitted for use as a lubricant for tablets and capsules at a final concentration of 1-3%; and iii) permitted for use as a slow release lipophilic matrix for sustained release tablets and capsules at a final concentration of 10-30% (up to a maximum of 500mg/person/d).



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			Glyceryl behenate is approved for use as an excipient ingredient for topical use at a concentration of 5% or less. In Listed Medicines: Behenic acid is a mandatory component of glyceryl behenate. When for oral ingestion, the maximum recommended daily dose must not provide more than 383.5 milligrams of behenic acid. In medicines for topical use, the concentration of glyceryl behenate must be no more than 5%.
Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Glyceryl dibehenate	<u>Role Non-medicinal:</u> Purposes: Lubricant , Skin-Conditioning Agent - Emollient Route Of Administration: Topical Allowed Purposes: Skin-Conditioning Agent - Emollient Reference: ICID Route Of Administration: Oral Allowed Purposes: Lubricant Reference: USP_31
Approved chemical name	Glyceryl behenate	<u>Role Non-medicinal:</u> Purposes: Emulsifying Agent , Lubricant , Skin-Conditioning Agent - Emollient
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE					
Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (Patch test)	Dermal	Human	COMPRITOL® 888 ATO 10% in Kernel Oil	Well tolerated Cutaneous irritation: 0.40	(Gattefossé data, GAT-96283, 1996)
Acute irritation (OECD 404)	Dermal	Rabbit	COMPRITOL® 888 ATO No dilution	Mildly Irritant	(Gattefossé data, GAT-99456, 1999)
Acute irritation (JORF)	Ocular	Rabbit	COMPRITOL® 888 ATO 5% in Kernel Oil	Slightly Irritant Ocular irritation (48h): 0.67	(Gattefossé data, GAT-95275, 1995)



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SAFETY STUDIES PERFORMED BY GATTEFOSSÉ

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (OECD 405)	Ocular	Rabbit	COMPRITOL® 888 ATO No dilution	Slightly Irritant	(Gattefossé data, GAT-99457, 1999)
Acute toxicity	Oral	Rat	COMPRITOL® 888 ATO Hydroxy cellulose gel 1%	LD _{0(oral)} > 5000 mg/kg	(Gattefossé data, GAT-80124, 1980)
Acute toxicity (OECD 401)	Oral	Rat	COMPRITOL® 888 ATO Suspended in Olive Oil	LD _{0(oral)} > 2000 mg/kg	(Gattefossé data, GAT-99458, 1999)

OECD : Organisation for Economic Co-operation and development - JORF: Journal officiel République Française (Official Journal French Republic)

Summary of files available upon request, please contact your Gattefossé local representative.
Additional literature review is available for the different endpoints, please consult the document "Tox and safety overview"
for a complete review on regulatory, tox and safety aspects.

CIR COMPENDIUM (Cosmetic Ingredient Review)

(<http://www.cir-safety.org>)

Denomination	Reference	Maximum "as used" concentration	Conclusions
Glyceryl Diesters (Glyceryl dibehenate)	IJT 26(S3) 2007	No information available	Based on the available animal and clinical data included in this report, the CIR Expert Panel concludes that the following Glyceryl diesters (glyceryl dibehenate ...) are safe as cosmetic ingredients in the practices of use and concentration as described in the safety assessment, provided that the content of 1,2-diester is not high enough to induce epidermal hyperplasia*.
Glyceryl monoesters (Glyceryl behenate)	IJT 23 (Suppl. 2: 55-94,) 2004	Glyceryl behenate: up to 5%	Based on the available animal and clinical data included in this report, the CIR Expert Panel concludes that the following Glyceryl Monoesters (glyceryl behenate ...) are safe as cosmetic ingredients in the present practices of use and concentration.

* 1,2-diester could potentially induce hyperplasia. The Panel did note that these compounds are more likely to cause these effects when the fatty acid chain length is ≤ 14 carbons, when one fatty acid is saturated and one is not, and when given at high doses, repeatedly.

Based on its composition, the available safety data and its history of use, we can conclude that for COMPRITOL® 888 ATO, the risk of epidermal hyperplasia is expected to be low and this ingredient is safe for use in cosmetic products (please contact us for additional info).



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9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
	CAS #	EC #	Chemical name
Glyceryl Behenate	77538-19-3 (or 91052-55-0) (or 30233-64-8) (or 94201-62-4) (or 18641-57-1)	278-717-5 (or 293-216-1) (or 250-097-0) (or 303-650-6) (or 242-471-7)	Docosanoic acid, ester with 1,2,3-propanetriol (or Glycerides, C16-22 mono-, di- and tri-) (or Docosanoic acid, monoester with 1,2,3-propanetriol) (or 2-hydroxypropane-1,3-diyl didocosanoate) (or Docosanoic acid, 1,2,3-propanetriyl ester)
EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list. EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5. Substances without EC numbers were assigned "list numbers" in order to facilitate REACH submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.			

NATIONAL INVENTORIES					
	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
Glyceryl Behenate	Not listed (Docosanoic acid, monoester with 1,2,3-propanetriol (CAS 30233-64-8) is listed)	NDSL listed (CAS 18641-57-1 is DSL listed)(R-ICL listed under Docosanoic acid, monoester with 1,2,3-propanetriol (CAS 30233-64-8) and Docosanoic acid, 2,3-dihydroxypropyl ester (CAS 6916-74-1))	Not listed (CAS 30233-64-8 and 18641-57-1 are listed)	Listed	Docosanoic acid, ester with 1,2,3-propanetriol (EINECS, IECSC, NDSL, NZIoC, REACH, TCSI VNECI) (or Glycerides, C16-22 mono-, di- and tri- (AREC, EINECS, IECSC, PICCS, REACH, TCSI, VNECI)) (or Docosanoic acid, monoester with 1,2,3-propanetriol (AIIC, AREC, NDSL, IECSC, NZIoC, REACH, TCSI, TSCA, VNECI)) (or 2-hydroxypropane-1,3-diyl didocosanoate (EINECS, IECSC, REACH, TCSI, VNECI)) (or Docosanoic acid, 1,2,3-propanetriyl ester (AIIC, DSL, EINECS, IECSC, INSQ, NZIoC, PICCS, REACH, TCSI, VNECI))
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting. Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation. Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					



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REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	Status per substance
Glyceryl Behenate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
GLYCERYL DIBEHENATE (UNII: R8WTH25YS2)	ORAL	CAPSULE	/	51 mg
	ORAL	CAPSULE, DELAYED RELEASE	3.5 mg	/
	ORAL	CAPSULE, EXTENDED RELEASE	/	1 mg
	ORAL	POWDER, FOR SOLUTION	/	8 mg
	ORAL	POWDER, FOR SUSPENSION	0.08 mg	/
	ORAL	SOLUTION	/	3 mg
	ORAL	TABLET	/	148 mg
	ORAL	TABLET, COATED	/	19 mg
	ORAL	TABLET, DELAYED RELEASE	/	75mg
	ORAL	TABLET, EXTENDED RELEASE	/	761mg
	ORAL	TABLET, FILM COATED	/	77 mg
	ORAL	TABLET, FILM COATED, EXTENDED RELEASE	/	10 mg
	ORAL	TABLET, ORALLY DISINTEGRATING	5.6 mg	/



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10.1.2. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENT (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	Functional Category	Regulatory status	Safety
Glyceryl Behenate	Coating agent; modified-release agent; tablet and capsule binder; tablet and capsule lubricant; taste-masking agent; viscosity-increasing agent.	GRAS listed. Accepted for use as a food additive in Europe. Included in the FDA Inactive Ingredients Database (oral capsules, tablets, and suspensions...). Included in the Canadian Natural Health Products Ingredients Database.	Glyceryl behenate is used in cosmetics, foods and oral pharmaceutical formulations, and is generally regarded as a relatively nonirritant and nontoxic material. The US Cosmetic Ingredients Review Expert Panel evaluated glyceryl behenate and concluded that it is safe for use in cosmetic formulations in present practices of use and concentration.

COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS		
Denomination	Category (concentration in %)	regulatory status
Glycerides of fatty acids	Suppository bases	1, 3–14, 16–26
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopeias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portuguese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian Garg et al., <i>Pharmaceutical Technology, Drug Delivery</i>, 2001, p14–24, Compendium Of Pharmaceutical Excipients For Vaginal Formulations		

10.2. Food/Nutraceutical/Dietary supplement applications

Mono- and Diglycerides and glyceryl behenate have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS471 Mono- and Diglycerides
E.U.	EFA (European Food additives)	E471 Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Mono- and Diglycerides Glyceryl behenate
U.S.A.	GRAS US Food additives according to 21 CFR	21 CFR §184.1505 Mono- and Diglycerides 21 CFR §184.1328 Glyceryl behenate
JAPAN	JSFA (Japanese Standards of Food Additives)	Glycerol esters of fatty acids

Remark: Compritol 888 ATO doesn't conform to monographs E471 and INS471 "Mono- and Diglycerides".

HANDBOOK of FOOD ADDITIVES		
Denomination	Use	Safety



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HANDBOOK of FOOD ADDITIVES

Compritol 888 Tribehenin	Emulsifier for foods; additive for tablet mfg.	FDA 21CFR §184.1328 GRAS; USP/NF, EP, and JSFA compliance; DMF no. 4663
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NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

Liability clause

The information contained in this document is based on our knowledge of the related product(s) at the date indicated herein, and is provided in good faith. Notwithstanding the foregoing, no guarantee is made in respect of the product's(s') properties or efficacy, or the continued exactitude and completeness of the information herein. No contractual tie is formed by the provision of this information. This document does not relieve the user from conducting all tests and verifications to ensure the product's(s') suitability for each intended use. Nothing herein shall be construed as a recommendation or licence to use any information that does or may conflict with any intellectual property. We make no representation or warranty, express or implied, that use of this information will not infringe any intellectual property of Gattefossé or any third party. The user assumes all risks and liability incurred if a product is used other than for the purpose designated by Gattefossé. This statement does not relieve the user from verifying its conformity with all current applicable laws and regulations, and the user shall remain solely liable for ensuring its continued conformity therewith.

